

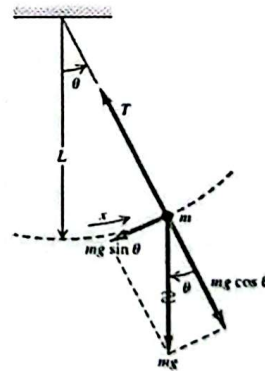
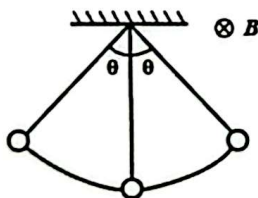
Simple Harmonic Motion

Periodic motion:

A motion which repeats itself over and over again after a regular interval of time is referred to as a periodic motion. The time required for each repetition is called time period. Example: the motion of moon about the earth, the motion of the piston of an engine, the oscillation of a pendulum, etc.

Oscillation:

When a particle, undergoing periodic motion, covers the same path back and forth about a mean position, it is said to be executing an oscillatory or vibratory motion. The oscillatory motion is therefore, a periodic “to and fro (forward and back ward)” motion. One complete “to and fro” motion is called an oscillation or vibration or a cycle. An oscillatory motion is not only periodic but also bounded, i.e., the displacement of the particle on either side of its mean position remains confined within a well defined limit. The number of complete oscillations or cycles in unit time is called the frequency of vibration. The displacement of a particle executing an oscillatory motion is bounded and usually expressed in terms of sines or cosines or combination of both



Simple Harmonic Motion (S.H.M):

The simplest type of periodic vibration is a simple harmonic motion, which is defined as the motion executed by a particle oscillating to and fro in a straight line under the action of a force which is proportional to the displacement of the particle from a fixed point and the direction of the force is always directed towards that point. Example: The motion of a simple pendulum.

Characteristic of S.H.M:

- i) The motion is periodic.
- ii) The motion takes place in straight line and its acceleration towards a fixed point.
- iii) Acceleration is proportional to its displacement from the mean position.

Displacement, Velocity, Acceleration and Time period in SHM

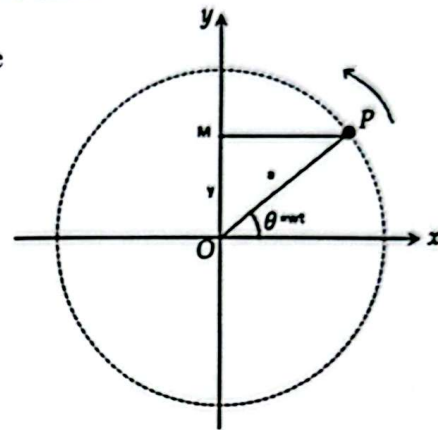
At any instant the displacement of M from centre O of the circle is called the displacement. If the particle moved from x to P in time t , then

$$\angle POX = \angle MPO = \theta = \omega t$$

From $\triangle MPO$,

$$\text{Since } \sin \theta = \sin \omega t = \frac{OM}{a}$$

$$\therefore \text{ displacement, } OM = y = a \sin \omega t$$



The maximum displacement of a vibrating particle is called its amplitude and the rate of change of displacement is called the velocity of the vibrating particle.

$$\begin{aligned} \therefore \text{ velocity} &= \frac{dy}{dt} = a\omega \cos \omega t \\ &= a\omega \sqrt{1 - \sin^2 \omega t} \\ &= a\omega \sqrt{1 - y^2 / a^2} \\ &= \omega \sqrt{a^2 - y^2} \end{aligned}$$

This equation clearly shows that the velocity of M is maximum when $y = 0$ and is equal to $a\omega$ and this is minimum when $y = a$ and is equal to 0. Hence, velocity of the particle M is a function of its displacement.

The rate of change of velocity of a vibrating particle is called its acceleration.

$$\begin{aligned} \therefore \text{ Acceleration} &= \text{rate of change velocity} \\ &= \frac{d}{dt} \left(\frac{dy}{dt} \right) \\ &= \frac{d^2 y}{dt^2} = -a\omega^2 \sin \omega t = -\omega^2 a \sin \omega t \end{aligned}$$

$$\therefore \frac{d^2y}{dt^2} = -w^2y$$

This equation clearly shows that acceleration of the particle is directly proportional to its displacement and negative sign indicates that it is acting towards the fixed point O, which is the main characteristic of S. H. M. Again if $y = 0$, acceleration = 0 (minimum);

$$y = a, \text{ acceleration} = -w^2a \text{ (maximum).}$$

The time period of the particle is defined as time taken by the particle to complete one revolution round the circle and hence is given by

$$T = \frac{2\pi}{w}$$

From

$$\frac{d^2y}{dt^2} = w^2y$$

Numerically,

$$w^2 = \frac{\text{acceleration}}{\text{displacement}}$$

$$\text{or, } w = 2\pi n = \sqrt{\frac{\text{acceleration}}{\text{displacement}}}$$

$$\text{or, } \frac{2\pi}{T} = \sqrt{\frac{\text{acceleration}}{\text{displacement}}}$$

$$\therefore T = 2\pi \sqrt{\frac{\text{displacement}}{\text{acceleration}}}$$

Equation for simple harmonic motion:

A simple harmonic motion may be defined as that motion in which the force acting on a particle and hence the acceleration of the particle at each instant is proportional to the negative of the displacement at that instant.

Thus, if F be the force acting on a particle executing SHM and y its displacement from its mean position or equilibrium then,

$$F = -ky$$

where, k = Proportionality constant

$$\text{or, } m \frac{d^2y}{dt^2} = -ky$$

$$\text{or, } \frac{d^2y}{dt^2} + \frac{k}{m}y = 0$$

$$w = \sqrt{\frac{k}{m}} = \text{angular velocity}$$

$$\text{or, } \frac{d^2 y}{dt^2} + w^2 y = 0 \quad \dots\dots\dots(1)$$

This is the differential equation for the SHM. To obtain the solution of equation (1), let us multiply it by $2 \frac{dy}{dt}$

$$\therefore 2 \frac{dy}{dt} \cdot \frac{d^2 y}{dt^2} + 2w^2 y \frac{dy}{dt} = 0$$

Integrating w. r. t. time

$$\left(\frac{dy}{dt} \right)^2 + w^2 y^2 = C$$

where, C is a constant of integration. C can be evaluated by applying boundary condition.

When the displacement of the particle is maximum, then $y = a$ (say) and $\frac{dy}{dt} = 0$ at that point.

Therefore, we get $C = w^2 a^2$

$$\text{or, } \left(\frac{dy}{dt} \right)^2 + w^2 y^2 = w^2 a^2$$

$$\text{or, } \left(\frac{dy}{dt} \right)^2 = w^2 (a^2 - y^2)$$

$$\text{or, } \frac{dy}{dt} = w \sqrt{a^2 - y^2}$$

$$\text{or, } \frac{dy}{\sqrt{a^2 - y^2}} = w dt$$

Integrating w. r. t. time again

$$\int \frac{dy}{\sqrt{a^2 - y^2}} = \int w dt$$

$$\text{or, } \sin^{-1} \frac{y}{a} = wt + \phi$$

$$\therefore y = a \sin(wt + \phi) \quad \dots\dots\dots(2)$$

This is the solution of a simple harmonic motion equation, which gives the displacement of a particle at any instant. Expanding equation (2)

$$\begin{aligned} y &= a \sin wt \cos \phi + a \cos wt \sin \phi \\ &= A \sin wt + B \cos wt \end{aligned}$$

where, $A = a \cos \phi$, $B = a \sin \phi$

In special cases (for $\phi = 0$ or $\frac{\pi}{2}$), the value A or B may be zero, the displacement may be written as either $y = A \sin \omega t$ or $y = B \cos \omega t$

Hence, the most general form of the differential equation is

$$y = A \sin \omega t + B \cos \omega t.$$

This is a combination of both the sine and the cosine terms.

Now from equation (2)

$$y = a \sin(\omega t + \phi)$$

$$\therefore \frac{dy}{dt} = v = a\omega \cos(\omega t + \phi)$$

$$\text{or, } \frac{d^2y}{dt^2} = -a\omega^2 \sin(\omega t + \phi)$$

$$\text{or, } \text{acceleration} = -\omega^2 y$$

$$\text{i.e., } \text{acceleration} \propto \text{displacement}$$

Again from equation (2)

$$y = a \sin(\omega t + \phi)$$

Let the time t in this equation is increased by $\frac{2\pi}{\omega}$, then

$$\begin{aligned} y &= a \sin \left[\omega \left(t + \frac{2\pi}{\omega} \right) + \phi \right] \\ &= a \sin(\omega t + 2\pi + \phi) \\ &= a \sin(\omega t + \phi) \end{aligned}$$

This is same as before. This indicates the time period of motion is $\frac{2\pi}{\omega}$

$$\therefore T = \frac{2\pi}{\omega}$$

$$\begin{aligned} \text{or, } T &= 2\pi \sqrt{\frac{1}{\omega^2}} = 2\pi \sqrt{\frac{m}{k}} \\ &= 2\pi \sqrt{\frac{1}{\mu}} \\ &= 2\pi \sqrt{\frac{1}{\text{acceleration per unit displacement}}} \end{aligned}$$

$$= 2\pi \sqrt{\frac{\text{displacement}}{\text{acceleration}}}$$

$$\therefore \text{frequency } n = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

We note that the time period of the motion does not depend on the amplitude of epoch or initial phase i.e., the oscillation is isochronous. This is an important property of SHM.

Energy of a particle executing SHM

The acceleration of a particle and hence the force acting on the particle executing simple harmonic motion is directed towards its mean position. Work is therefore done during the displacement of the particle. Hence the particle possesses potential energy (U). As the particle also possesses velocity, it possesses kinetic energy (K) too. Thus the total energy of a particle is partly kinetic and partly potential. We know the displacement of a particle executing SHM is

$$y = a \sin(\omega t + \phi) \quad \text{where, } y = \text{displacement, } a = \text{amplitude}$$

$$\therefore v = \frac{dy}{dt} = a\omega \cos(\omega t + \phi) \quad \omega = \text{angular velocity, } \phi = \text{epoch}$$

$$\begin{aligned} \therefore \text{Kinetic energy, KE} &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}ma^2\omega^2 \cos^2(\omega t + \phi) \\ &= \frac{1}{2}ka^2 \cos^2(\omega t + \phi) \end{aligned}$$

The potential energy is the amount of work that must be done in overcoming the force through the displacement.

$$\begin{aligned} \therefore \text{Potential energy, PE} &= \int_0^y F dy \\ &= \int_0^y ky dy = \frac{1}{2}ky^2 \\ &= \frac{1}{2}m\omega^2 a^2 \sin^2(\omega t + \phi) \\ &= \frac{1}{2}ka^2 \sin^2(\omega t + \phi) \end{aligned}$$

$$\therefore \text{Total energy} = \text{KE} + \text{PE}$$

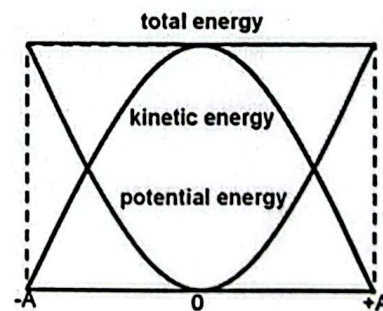
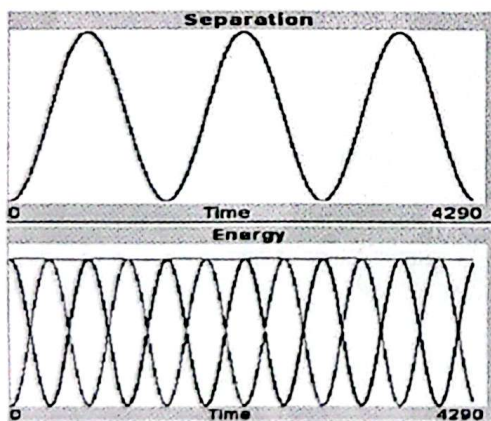
$$\begin{aligned}
&= \frac{1}{2}ka^2[\cos^2(\omega t + \phi) + \sin^2(\omega t + \phi)] \\
&= \frac{1}{2}ka^2 = \frac{1}{2}m\omega^2a^2 = \frac{1}{2}m\left(\frac{2\pi}{T}\right)^2a^2 \\
&= 2\pi^2n^2a^2m \text{ where, } \frac{1}{T} = n \rightarrow \text{frequency of oscillation}
\end{aligned}$$

$$\text{At } t = 0, \phi = 0, \quad KE_{\max} = \frac{1}{2}ka^2$$

$$\text{At } t = 0, \phi = \frac{\pi}{2}, \quad PE_{\max} = \frac{1}{2}ka^2$$

We see that, the total energy = $KE_{\max} = PE_{\max}$ i.e., the total energy of the system is same as the maximum value of any one of the two forms of energy. At the maximum displacement the KE is zero but the PE is maximum, has the value $\frac{1}{2}ka^2$. At the equilibrium position the PE is zero and the KE is maximum, has the value $\frac{1}{2}ka^2$. At any other position the energy is half kinetic and half potential or sum is always $\frac{1}{2}ka^2$.

Figure shows the kinetic energy, potential energy and total energy as a function of time and position.



Average value of kinetic and potential energies of a SHM

The KE of the particle executing SHM is given by

$$KE = \frac{1}{2} m \omega^2 a^2 \cos^2(\omega t + \phi)$$

The average KE over a complete cycle or a whole time period is

$$\begin{aligned} K_{avg} &= \frac{1}{T} \int_0^T \frac{1}{2} m \omega^2 a^2 \cos^2(\omega t + \phi) dt \\ &= \frac{1}{T} \frac{m \omega^2 a^2}{4} \int_0^T 2 \cos^2(\omega t + \phi) dt \\ &= \frac{m \omega^2 a^2}{4T} \int_0^T [1 + \cos 2(\omega t + \phi)] dt \\ &= \frac{m \omega^2 a^2}{4T} \left[\int_0^T dt + \int_0^T \cos 2(\omega t + \phi) dt \right] \end{aligned}$$

The average value of both a sine and a cosine function for a complete cycle or a whole time period T is zero. Hence

$$\begin{aligned} K_{avg} &= \frac{m \omega^2 a^2}{4T} \times T + (\text{the last term is zero}) \\ &= \frac{1}{4} m \omega^2 a^2 = \frac{1}{4} k a^2 \\ &= \frac{1}{4} m (2\pi n)^2 a^2 = \pi^2 n^2 a^2 m \end{aligned}$$

Similarly average PE of the particle over a complete cycle is

$$\begin{aligned} U_{avg} &= \frac{1}{T} \int_0^T \frac{1}{2} m \omega^2 a^2 \sin^2(\omega t + \phi) dt \\ U_{avg} &= \frac{m \omega^2 a^2}{4T} \int_0^T 2 \sin^2(\omega t + \phi) dt \\ &= \frac{m \omega^2 a^2}{4T} \int_0^T [1 - \cos 2(\omega t + \phi)] dt \\ &= \frac{m \omega^2 a^2}{4T} \times T - 0 \\ &= \frac{1}{4} m \omega^2 a^2 = \frac{1}{4} k a^2 \\ &= \pi^2 n^2 a^2 m \end{aligned}$$

Thus, average value of KE of the particle = average value of PE of the particle = $\frac{1}{4}ka^2$ = half the total energy.

Examples of SHM:

(a) The simple pendulum

It consists of a point mass suspended by an in extensible weight less string in a uniform gravitational field. When pulled to one side of its equilibrium position and released, the bob of the pendulum vibrates about the position of equilibrium. Let a pendulum be displaced from its mean position O and allowed to oscillate. The force on the bob is mg acting down ward.

Resolving mg

- i) $mg \cos \theta \rightarrow$ along string balanced by tension T
- ii) $mg \sin \theta \rightarrow$ perpendicular to the string which acting to oscillate the bob.

The restoring force F is

$$F = -mg \sin \theta \quad [-\text{ve sign shows that acceleration directed towards mean position}]$$

or, $F = -mg \theta$

or, $m \frac{d^2 y}{dt^2} = -mg \frac{y}{l}$ where, $\theta = \frac{y}{l}$, $l \rightarrow$ length of the string, $y \rightarrow$ displacement

or, $\frac{d^2 y}{dt^2} + \frac{g}{l} y = 0$

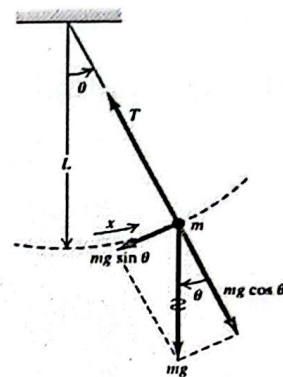
or, $\frac{d^2 y}{dt^2} + w^2 y = 0$ where, $w \rightarrow \sqrt{\frac{g}{l}}$

This is the differential equation for SHM. The time period of oscillation of the simple pendulum is

$$T = \frac{2\pi}{w} = 2\pi \sqrt{\frac{l}{g}}$$

Draw backs of simple pendulum

- i) It is an ideal conception not realized in practice
- ii) The resistance and buoyancy of the air affect the motion
- iii) The expression for time period is true only for infinitely small amplitude



- iv) The motion of the bob is not strictly linear
- iv) The bob has also relative motion with respect to string.

(b) Compound pendulum

A compound pendulum is a rigid mass capable of oscillating about a horizontal axis passing through any point of the mass. In figure, C is the centre of gravity of the body and S is the point of the suspension.

At any instant of time, when the mass has been displaced, the force acting vertically downwards = mg . At this position, the line SC makes an angle θ with the vertical axis. The restoring moment of this forces about the point S is $mg l \sin \theta$.

Let I be the moment of inertial of the pendulum about an axis passing through S. Then,

$$I \frac{d^2 \theta}{dt^2} = -mgl \sin \theta \quad [-ve \text{ sign shows that the force is directed towards mean position}]$$

$$\text{or,} \quad m(K^2 + l^2) \frac{d^2 \theta}{dt^2} = -mgl \sin \theta$$

where, K = radius of gyration about the axis passing through the centre of gravity.

$$\therefore \quad m(K^2 + l^2) \frac{d^2 \theta}{dt^2} + mgl \theta = 0 \quad [\text{for } \theta \text{ is small}]$$

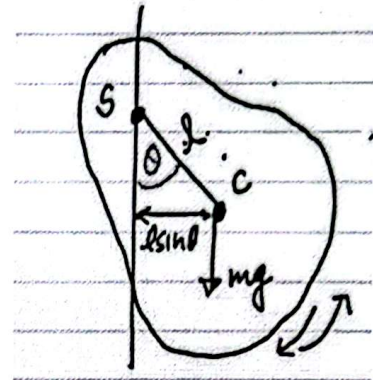
$$\text{or,} \quad \frac{d^2 \theta}{dt^2} + \left(\frac{gl}{K^2 + l^2} \right) \theta = 0$$

$$\text{or,} \quad \frac{d^2 \theta}{dt^2} + w^2 \theta = 0$$

This is the differential equation of SHM. The time period of a compound pendulum is

$$\begin{aligned} T &= 2\pi \sqrt{\frac{1}{w}} = 2\pi \sqrt{\frac{K^2 + l^2}{gl}} \\ &= 2\pi \sqrt{\frac{K^2 / l + l}{g}} \end{aligned}$$

Here, $K^2 / l + l$ is the equivalent length of the simple pendulum.



Superiority of compound pendulum over simple pendulum

- (i) A compound pendulum is easily realized in actual practice
- (ii) It oscillates as a whole and there is lag like that between the bob and the string in the case of simple pendulum.
- (iii) On account of its large mass and hence a large moment of inertia, it continues to oscillate for a longer time, so time period calculated is more accurate.